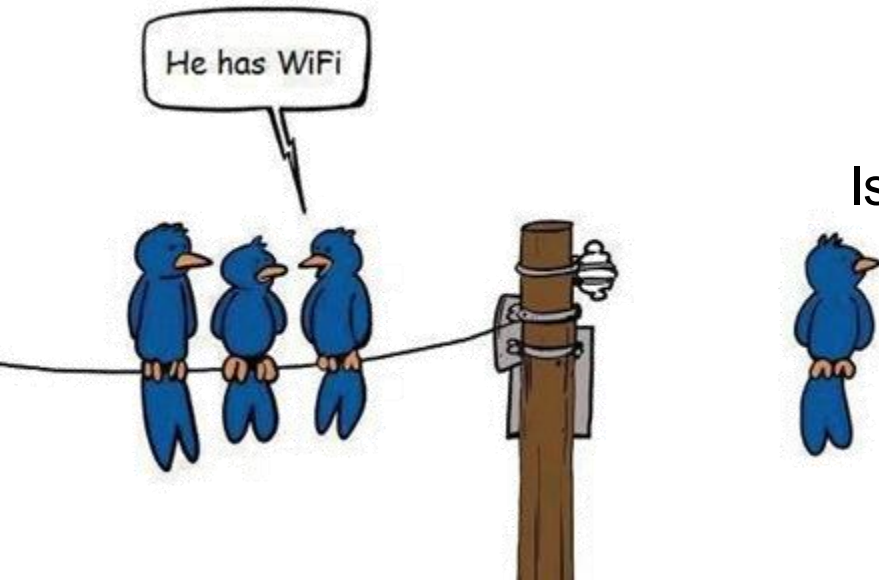


ECSE 4560/ 6560

Modern Communication Systems

Lecture 16: MIMO Intro

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **MIMO**
- Projects
 - Group formation [MCS_Fall25_project.xlsx](#)
 - Project mid-term report due **Friday Nov 7**
 - Final Project Presentation and Demos: Dec 9 and Dec 12
 - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
 - HW/Lab 4 due Mon Oct 27 (Late due Wed **Oct 29**). HW5 coming soon
 - Exam 1: Friday Oct 24
 - Exam 2: **Friday Nov 21**

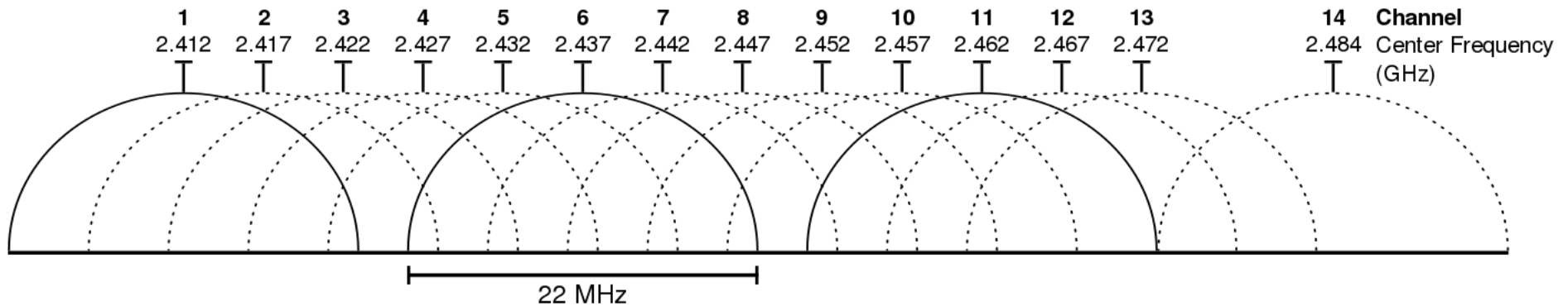
Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - MIMO Chapter 6
- An Adaptive Orthogonal Frequency Division Multiplexing Baseband Modem for Wideband Wireless Channels; by Jit Ken Tan, MIT Thesis
 - OFDM Decoding and Synchronization
 - Lab 4 Implementation

Today's Class

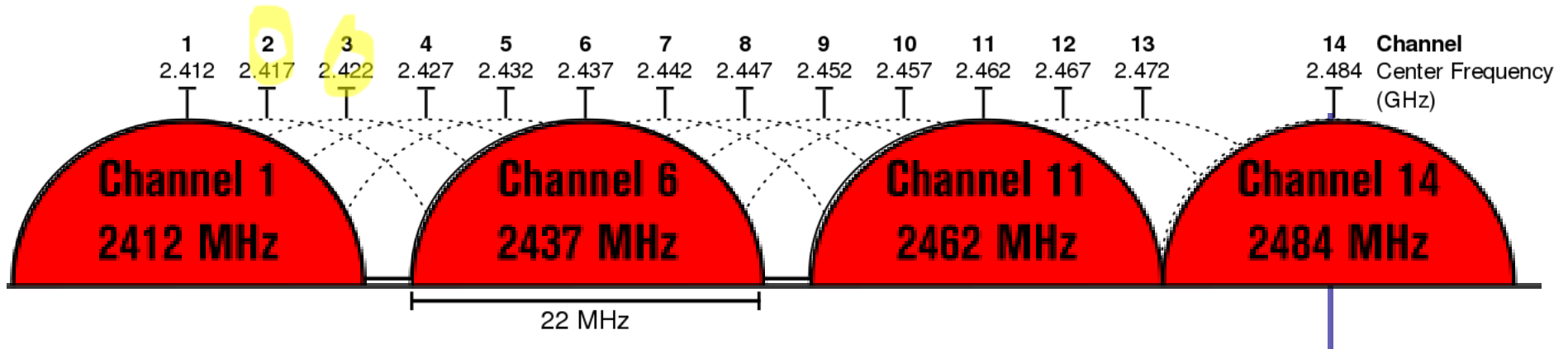
- **MIMO Intro**
 - **MIMO in WiFi Standards**
 - **MIMO Intro**
 - **MIMO antenna-space vector representation**
- **MIMO Channel**
 - Channel representation
 - Preamble based channel estimation
- **MIMO Gains**
 - Receiver Diversity
 - Transmitter Diversity

WiFi Standards from WiFi 1 to WiFi 6



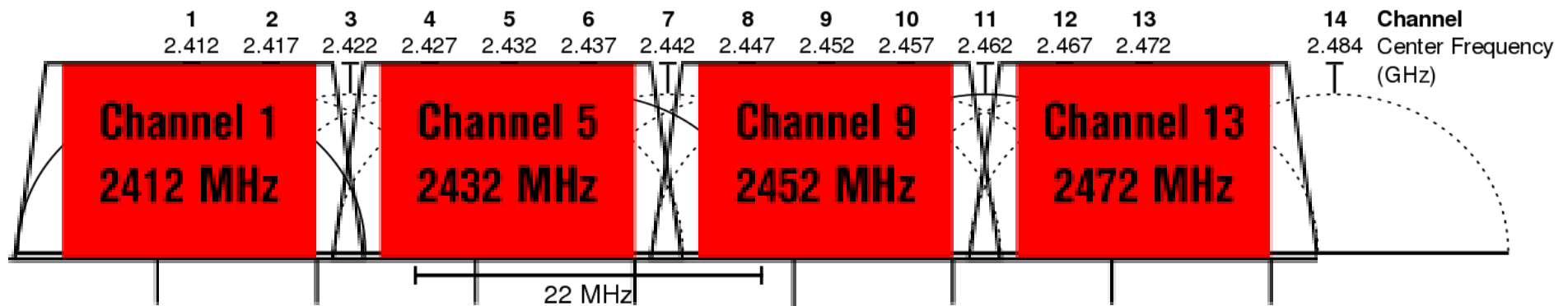
Version	Year	Technology	Modulation	Freq.	Bandwidth	Maximum Data Rate
WiFi 1 (802.11b)	1999	DSSS	DBPSK, DQPSK,	2.4 GHz	22 MHz	11 Mb/s

WiFi Standards from WiFi 1 to WiFi 6



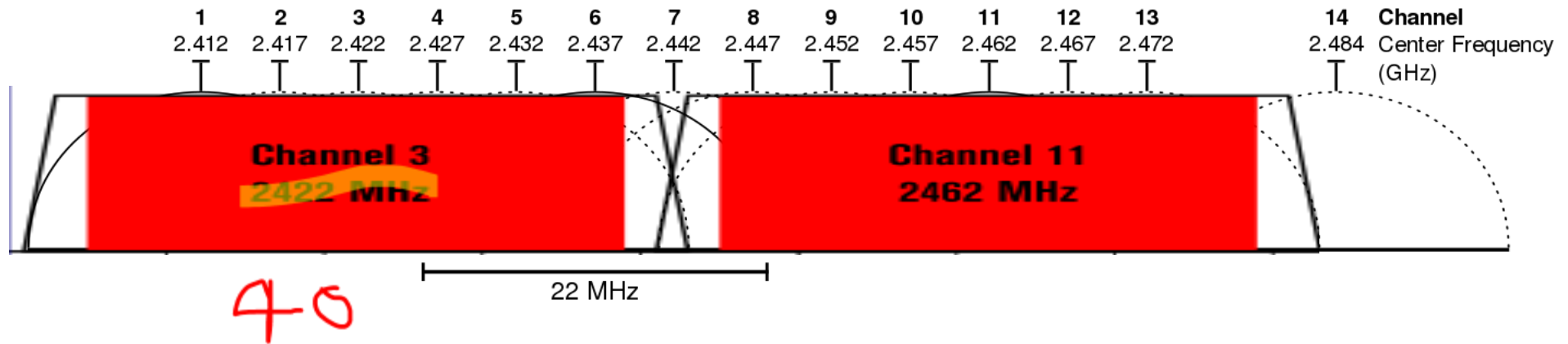
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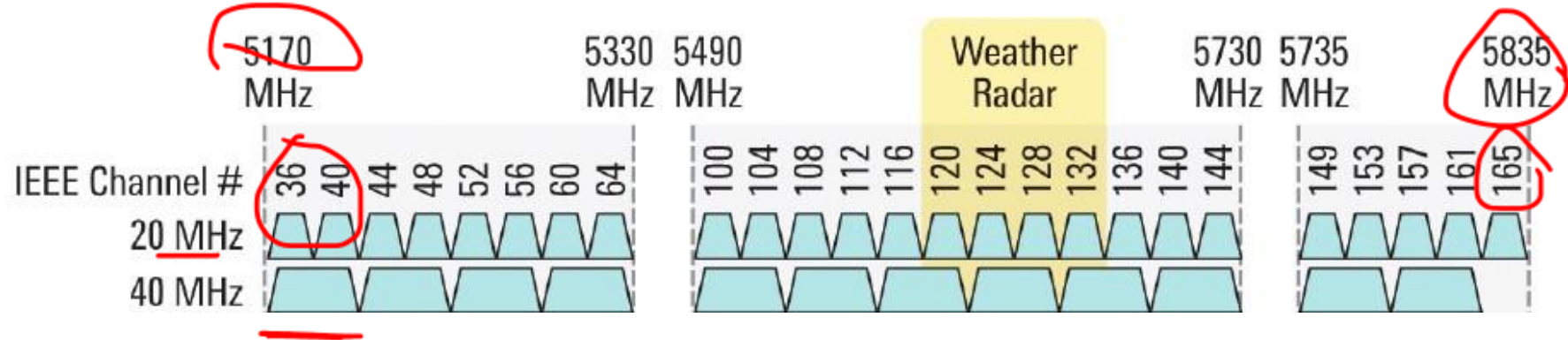
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WiFi Standards from WiFi 1 to WiFi 6



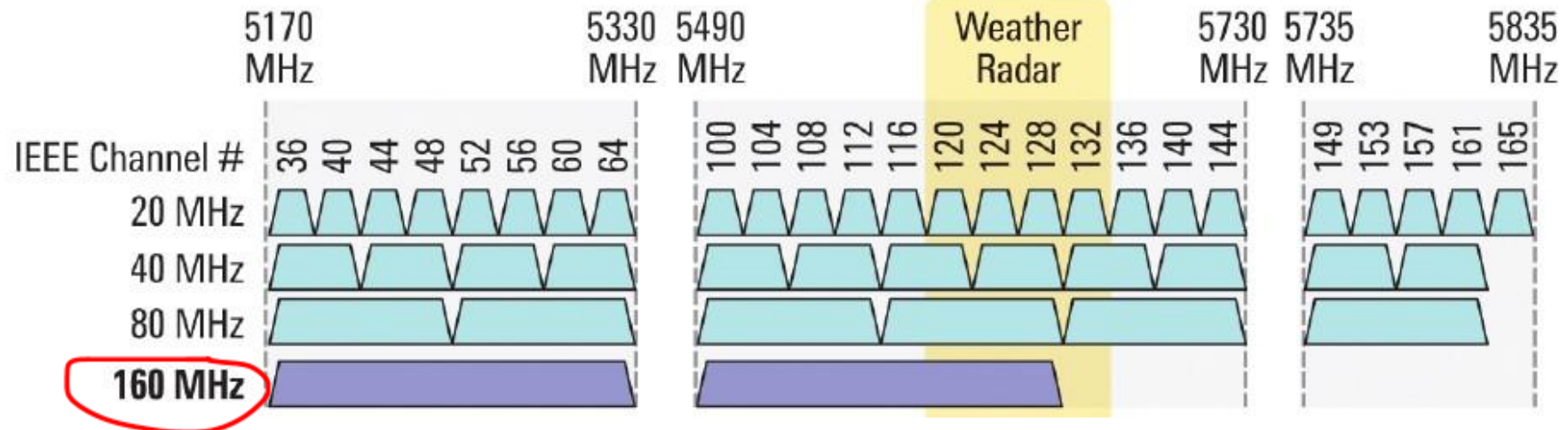
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WiFi Standards from WiFi 1 to WiFi 6



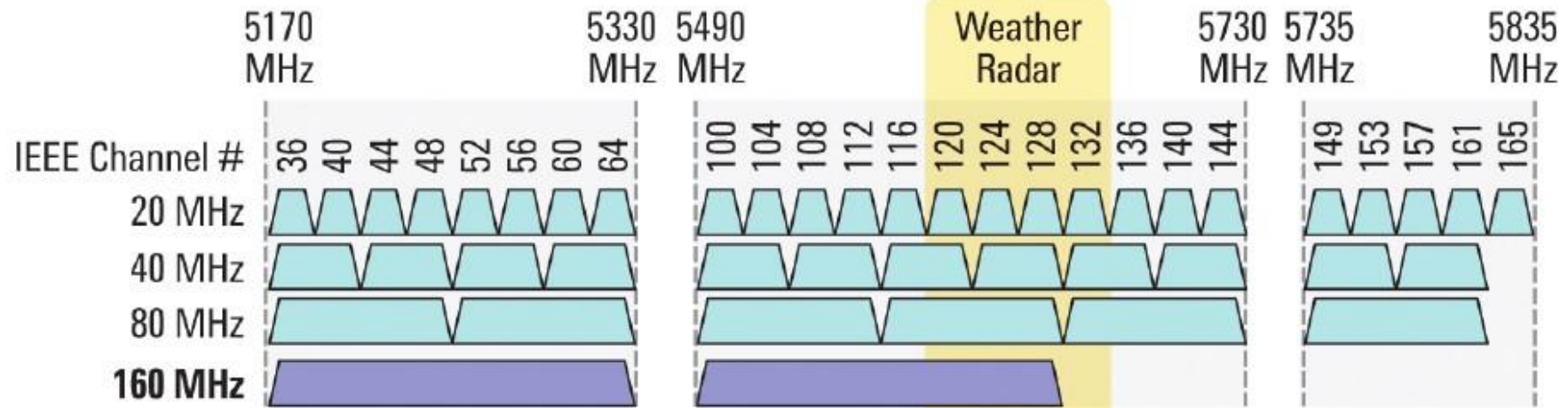
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WiFi Standards from WiFi 1 to WiFi 6



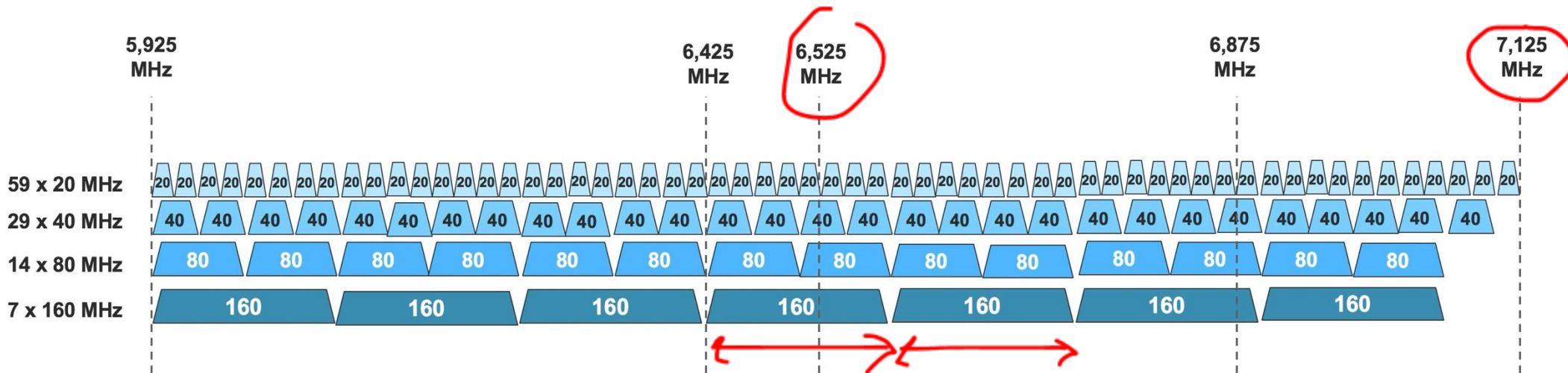
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WiFi 5 (802.11ac)	2014	OFDM (N=64, 128, 256, <u>512</u>) MIMO (8x8) <u>MU-MIMO (Downlink)</u>	BPSK, QPSK, 16-QAM, 64-QAM <u>256-QAM</u>	5 GHz	20 MHz 40 MHz 80 MHz <u>160 MHz</u>	<u>6.933 Gb/s</u>

WiFi Standards from WiFi 1 to WiFi 6



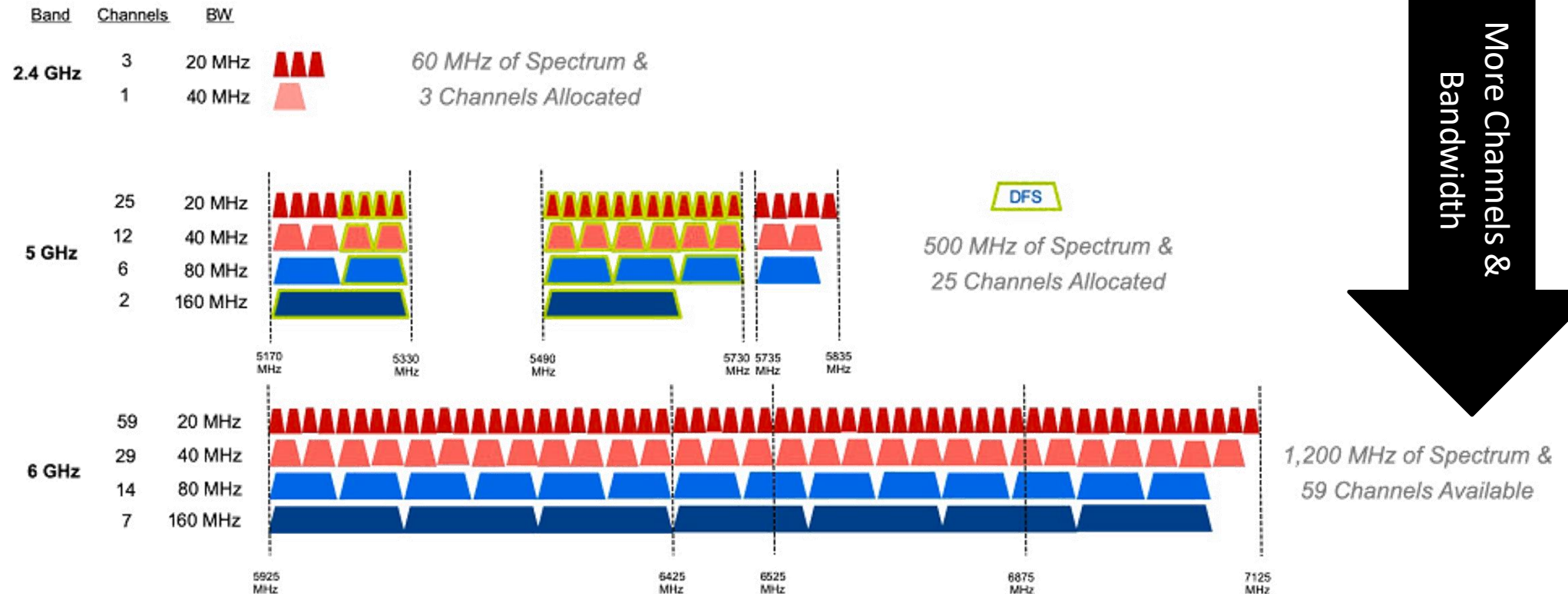
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WiFi 5 (802.11ac)	2014	OFDM (N=64, 128, 256, 512) MIMO (8x8) MU-MIMO (<u>Downlink</u>)	BPSK, QPSK, 16-QAM, 64-QAM 256-QAM	5 GHz	20 MHz 40 MHz 80 MHz 160 MHz	6.933 Gb/s
<u>WiFi 6 (802.11ax)</u>	<u>2019</u>	OFDM (N=256, 512, 1024, <u>2048</u>) MIMO (<u>8x8</u>) MU-MIMO (<u>Up & Down</u>) <u>OFDMA</u>	BPSK, QPSK, 16-QAM, 64-QAM 256-QAM, 1024-QAM	2.4 GHz 5 GHz	20 MHz 40 MHz 80 MHz <u>160 MHz</u>	<u>9.608 Gb/s</u>

WiFi Standards from WiFi 1 to WiFi 6



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WiFi 6 (802.11ax) WiFi 6E	2020	<u>OFDM</u> (N=256, 512, 1024, 2048) MIMO (8x8) MU-MIMO (Up & Down) OFDMA	BPSK, QPSK, 16-QAM, 64-QAM 256-QAM, 1024-QAM	2.4 GHz 5 GHz 6 GHz	20 MHz 40 MHz 80 MHz 160 MHz	<u>9.608 Gb/s</u>

WiFi Standards from WiFi 1 to WiFi 6



More Channels & Bandwidth



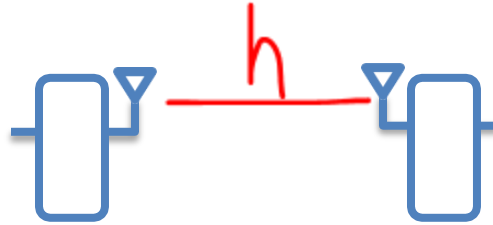
WiFi 1

More MIMO Antennas

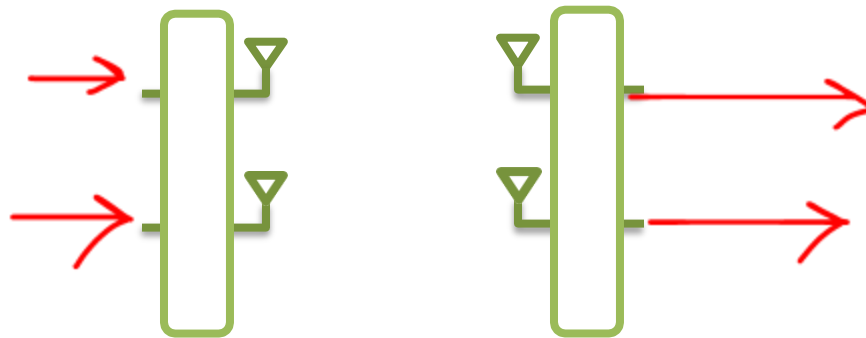
WiFi 5/6

MIMO: Multiple Input Multiple Output

So far: single input single output



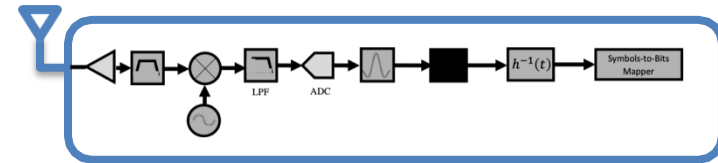
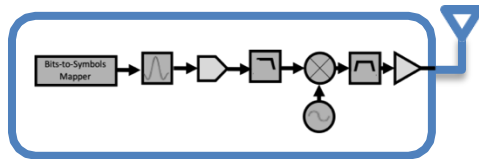
MIMO: multiple input multiple output



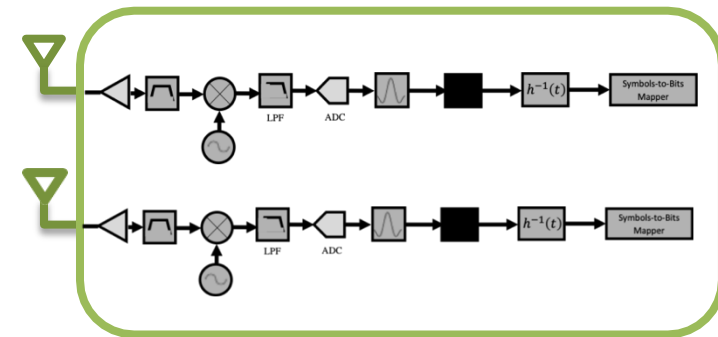
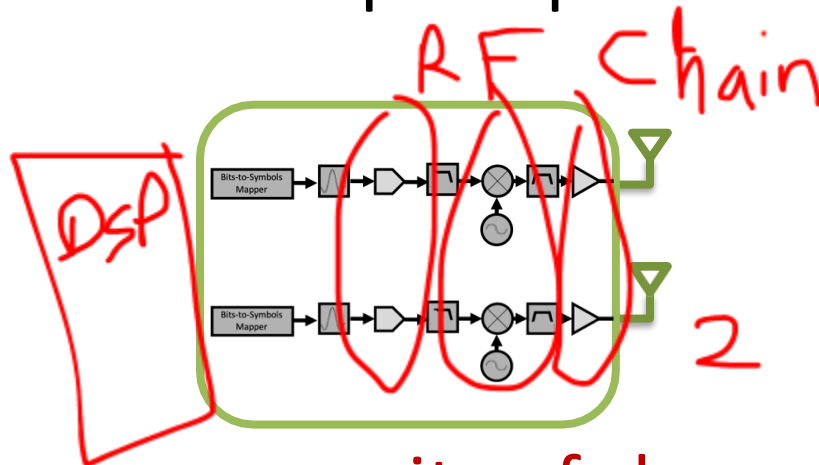
Increase capacity of channel using multiple transmit
and receive antennas.

MIMO: Multiple Input Multiple Output

So far: single input single output

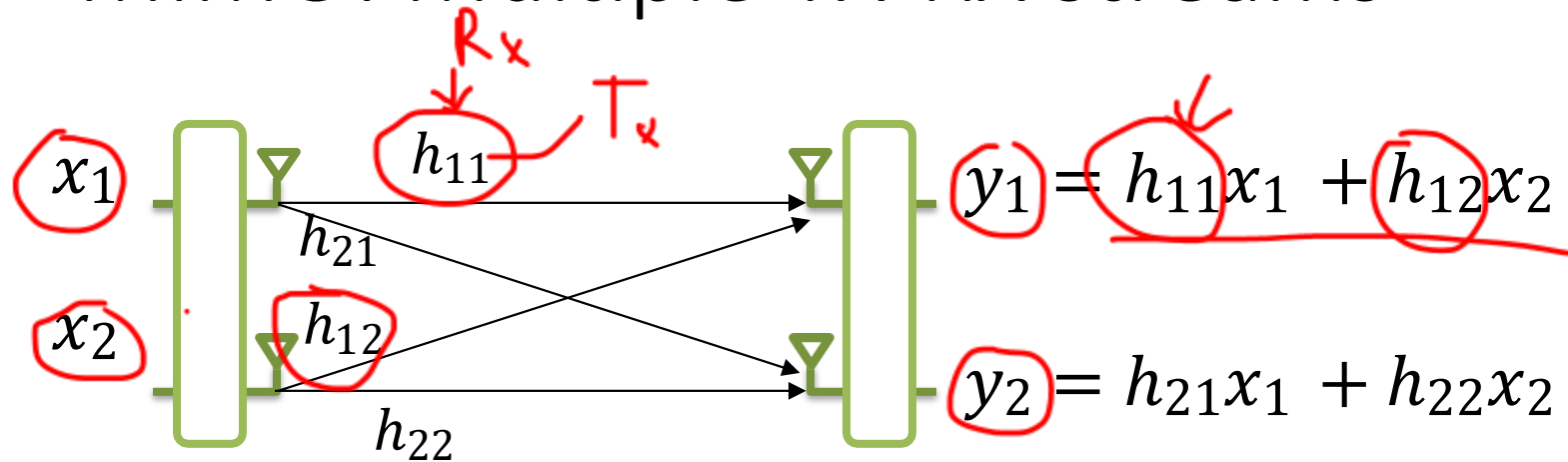


MIMO: multiple input multiple output



Increase capacity of channel using multiple transmit and receive antennas.

MIMO: Multiple TX-RX streams



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \rightarrow \boxed{\bar{\mathbf{y}} = \mathbf{H}\bar{\mathbf{x}} + \bar{\mathbf{n}}}$$

Handwritten red annotations: $\mathbf{H}$ is written below the channel matrix, and "noise" is written below the noise vector $\begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$. An arrow points from the boxed equation back to the noise vector.

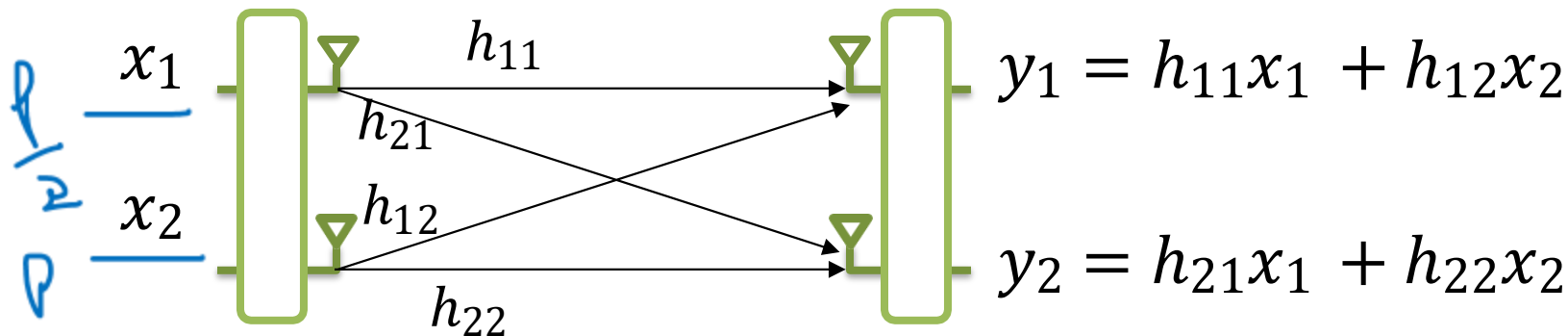
How to recover x_1 and x_2 ?

Estimate $\mathbf{H}$, compute $\mathbf{H}^{-1}$ and invert the channel!

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Transmit 2 packets at the same time!

MIMO: Multiple TX-RX streams



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

$$\Rightarrow \mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

How to recover x_1 and x_2 ?

For N antennas, $\mathbf{H}$ is $N \times N$ matrix $\rightarrow O(N^3)$

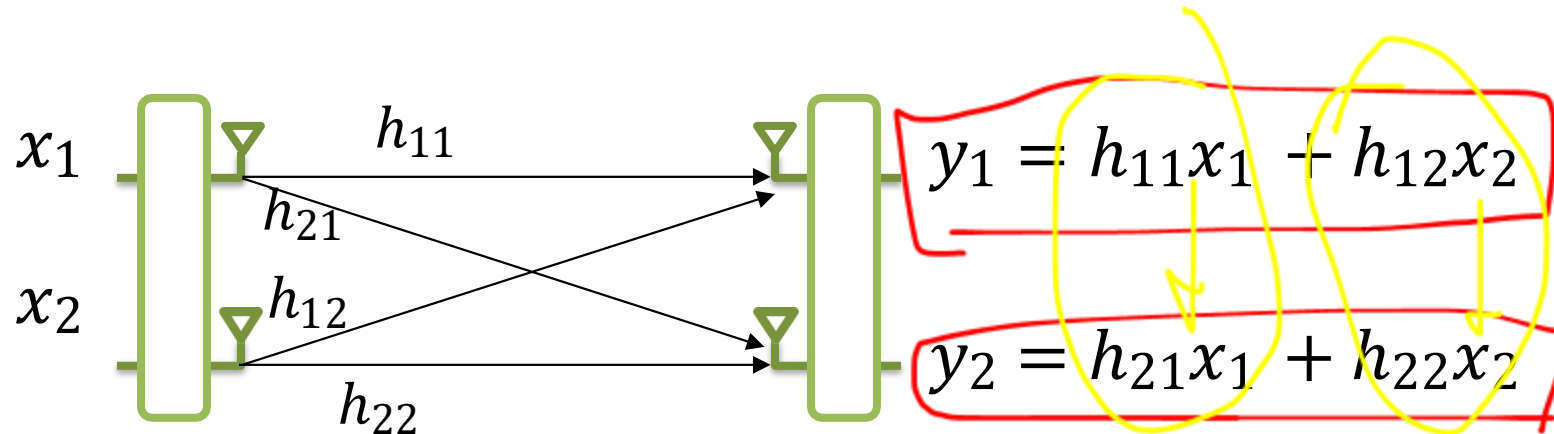
Estimate $\mathbf{H}$, compute $\mathbf{H}^{-1}$ and invert the channel!

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

SVD

Noise amplification

MIMO: Vector Representation



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \quad \Rightarrow \quad \mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

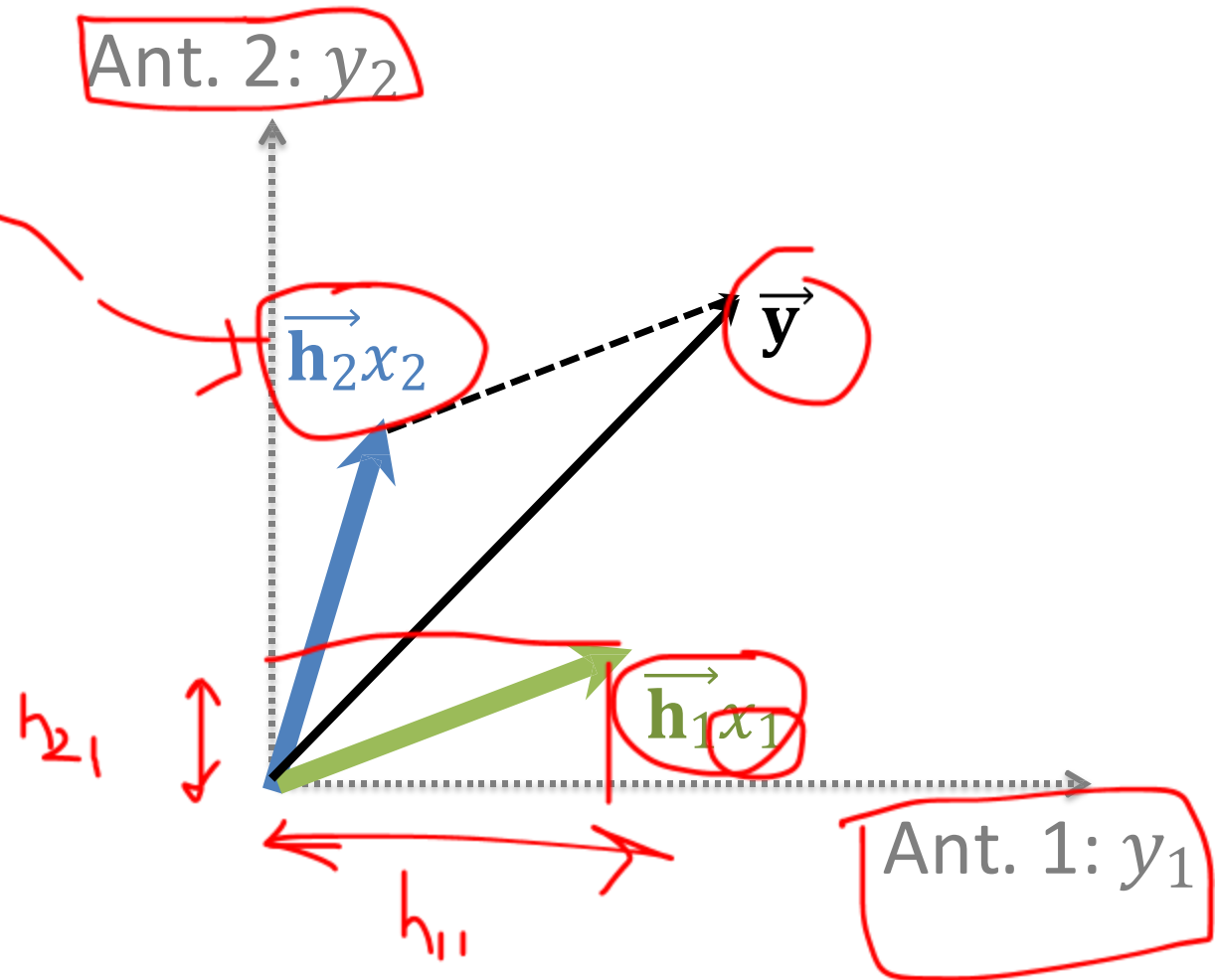
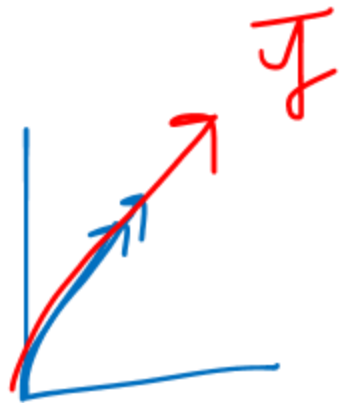
How to recover x_1 and x_2 ?

$$\underline{\mathbf{y}} = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix} x_1 + \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} x_2 = \underline{\vec{\mathbf{h}}_1} x_1 + \underline{\vec{\mathbf{h}}_2} x_2$$

2.

MIMO: Antenna Space

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2$$



$\vec{h}_1 = \vec{h}_2$ can't recover x_1 & x_2

MIMO: Antenna Space

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2$$

To decode x_1 , project on a vector $\vec{h}_2^\perp$ orthogonal to $\vec{h}_2$

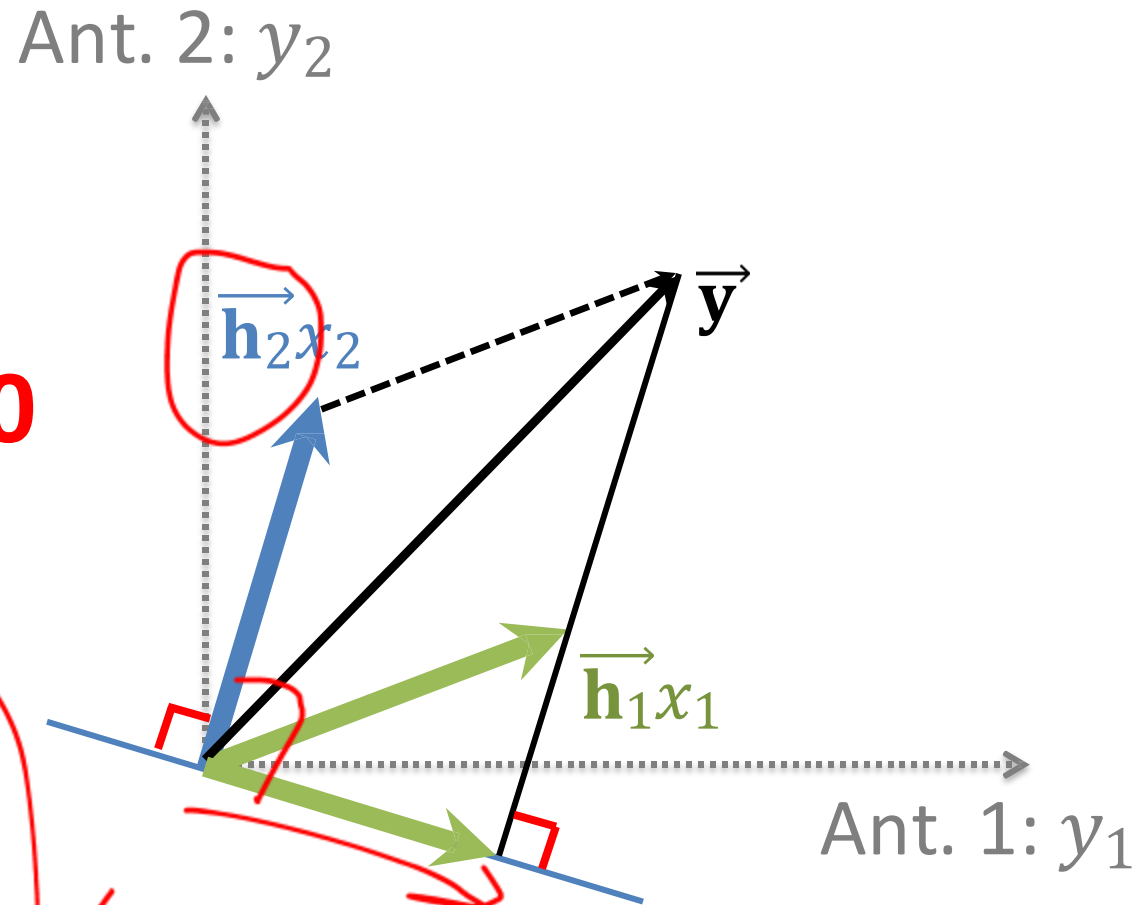
$$\vec{h}_2^\perp \vec{y} = \vec{h}_2^\perp \vec{h}_1 x_1 + \vec{h}_2^\perp \vec{h}_2 x_2$$

$$= \vec{h}_2^\perp \vec{h}_1 x_1$$

$$\vec{y} = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix} x_1 + \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} x_2$$

$$\vec{h}_2^\perp = \begin{bmatrix} h_{22} & -h_{12} \end{bmatrix}$$

$$\begin{aligned} \vec{h}_2^\perp \vec{y} &= h_{22} h_{11} x_1 - h_{12} h_{21} x_1 \\ &\quad + \cancel{h_{22} h_{12} x_2} - \cancel{h_{12} h_{22} x_2} \\ &= (h_{22} h_{11} - h_{12} h_{21}) x_1 \end{aligned}$$



MIMO: Antenna Space

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2$$

To decode x_2 , project on a vector $\vec{h}_1^\perp$ orthogonal to $\vec{h}_1$

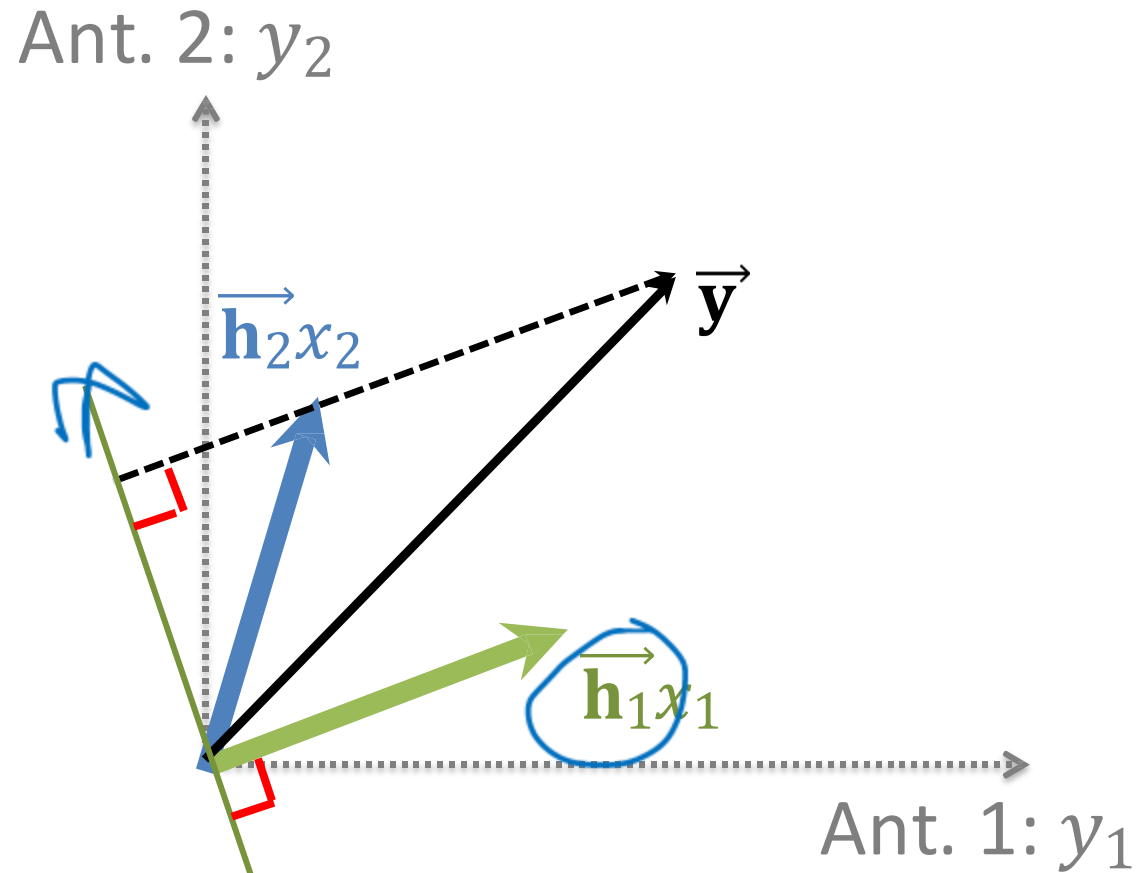
$$\vec{h}_1^\perp \vec{y} = \vec{h}_1^\perp \vec{h}_1 x_1 + \vec{h}_1^\perp \vec{h}_2 x_2$$

$$= \vec{h}_1^\perp \vec{h}_2 x_2$$

$$\mathbf{y} = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix} x_1 + \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} x_2$$

$$\vec{h}_1^\perp = \begin{bmatrix} h_{21} & -h_{11} \end{bmatrix}$$

$$\begin{aligned} \vec{h}_1^\perp \vec{y} &= \cancel{h_{21} h_{11} x_1} - \cancel{h_{21} h_{11} x_1} + h_{21} h_{12} x_2 - h_{11} h_{22} x_2 \\ &= (h_{21} h_{12} - h_{11} h_{22}) x_2 \end{aligned}$$

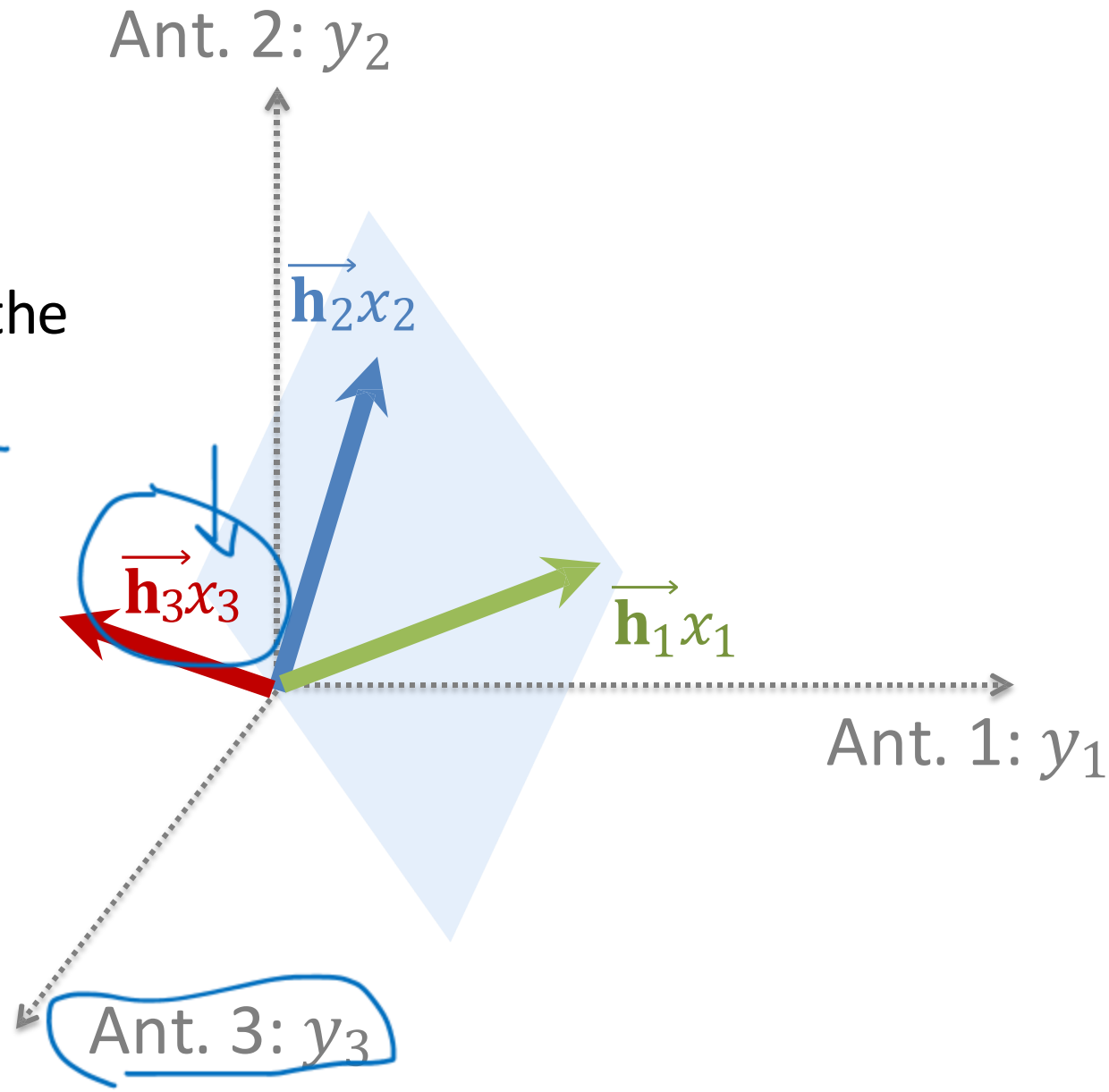
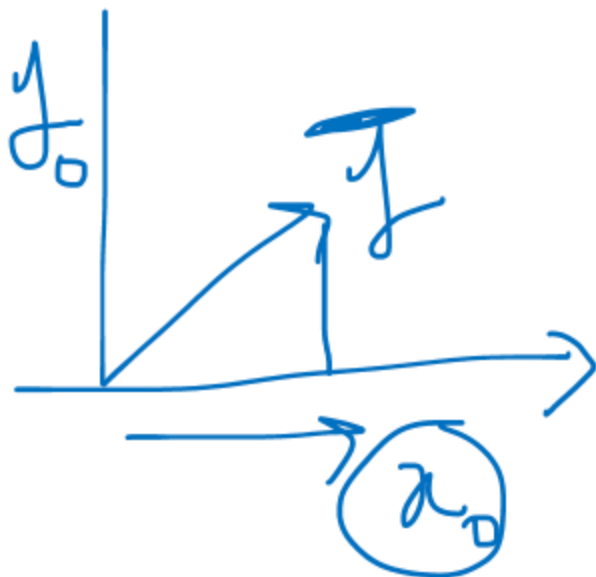


MIMO: Antenna Space

What about MIMO with more antennas?

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2 + \vec{h}_3 x_3$$

To decode x_1 , project on a vector $\vec{\mathbf{h}}_{23}$ $\perp$ orthogonal to the plane formed by $\vec{\mathbf{h}}_2$ and $\vec{\mathbf{h}}_3$

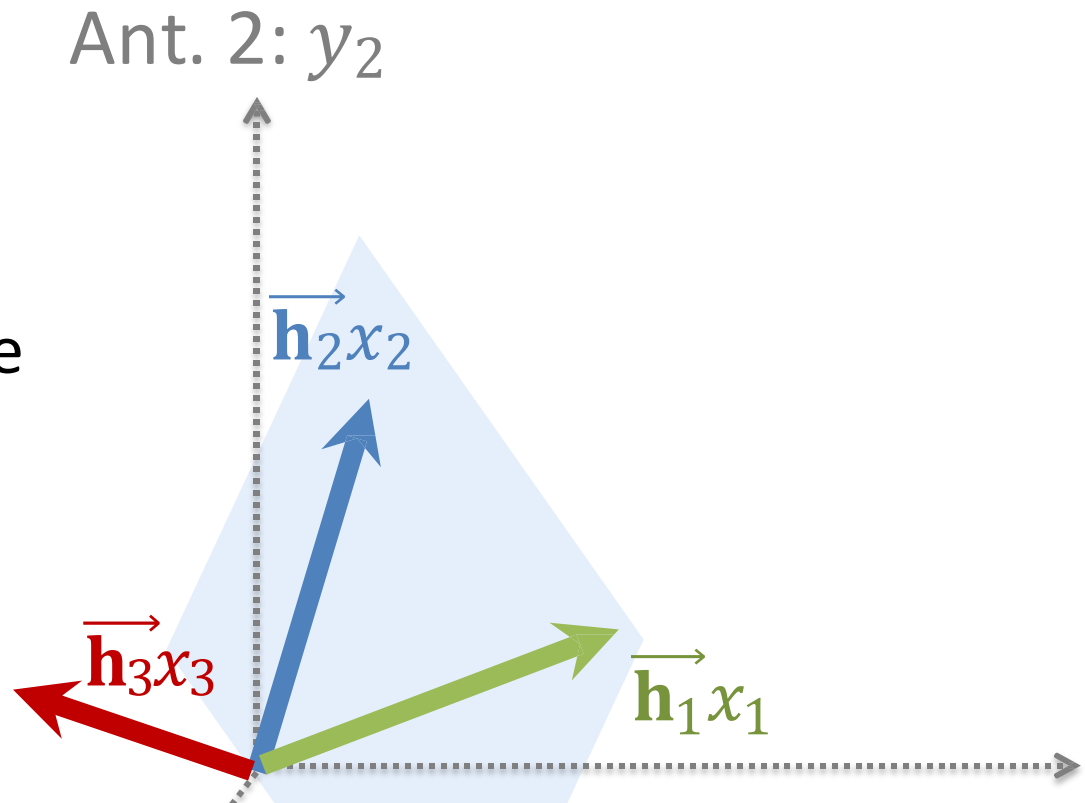


MIMO: Antenna Space

What about MIMO with more antennas?

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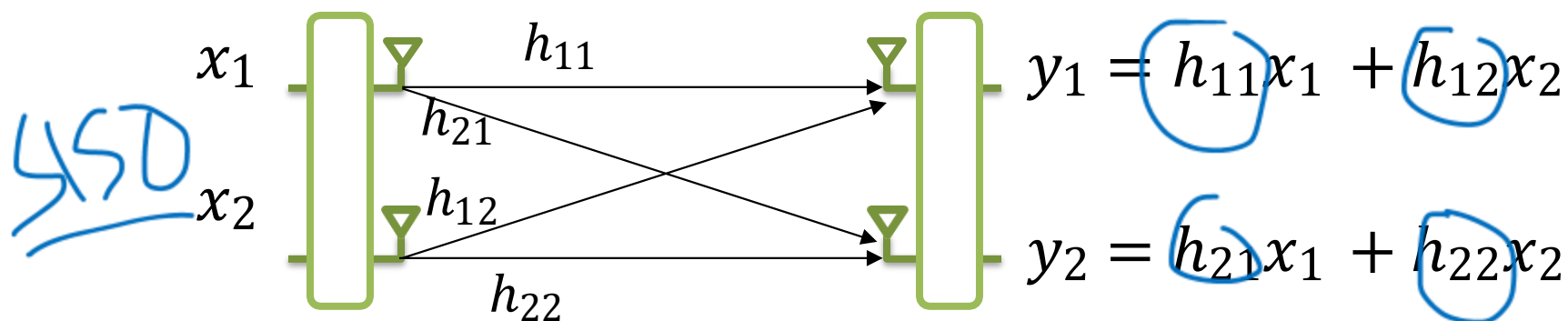


N antenna MIMO, receives signals in N dimensional space
→ Can decode N parallel signals

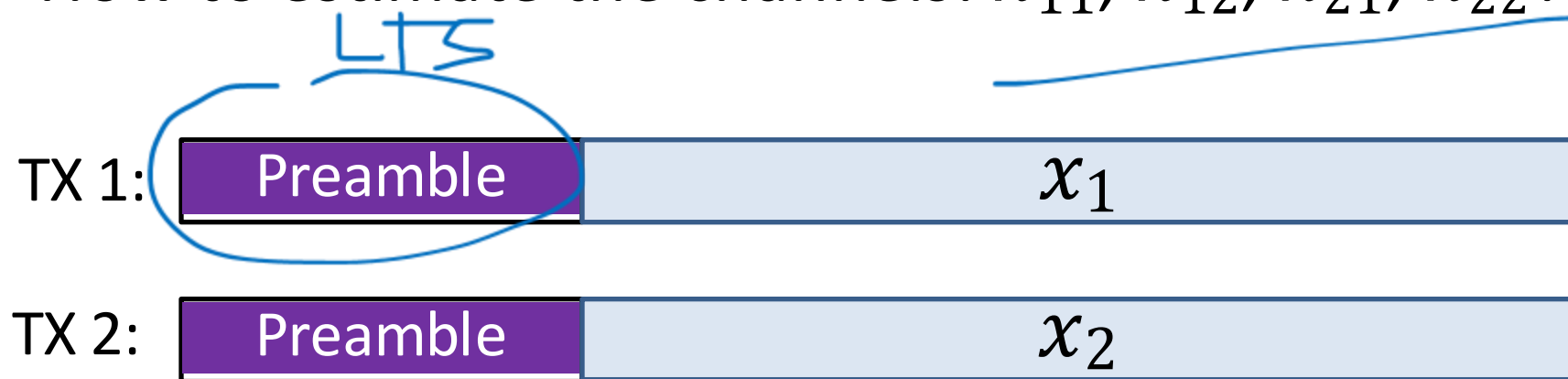
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- **MIMO Channel**
 - **Channel representation**
 - **Preamble based channel estimation**
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 - Transmitter Diversity

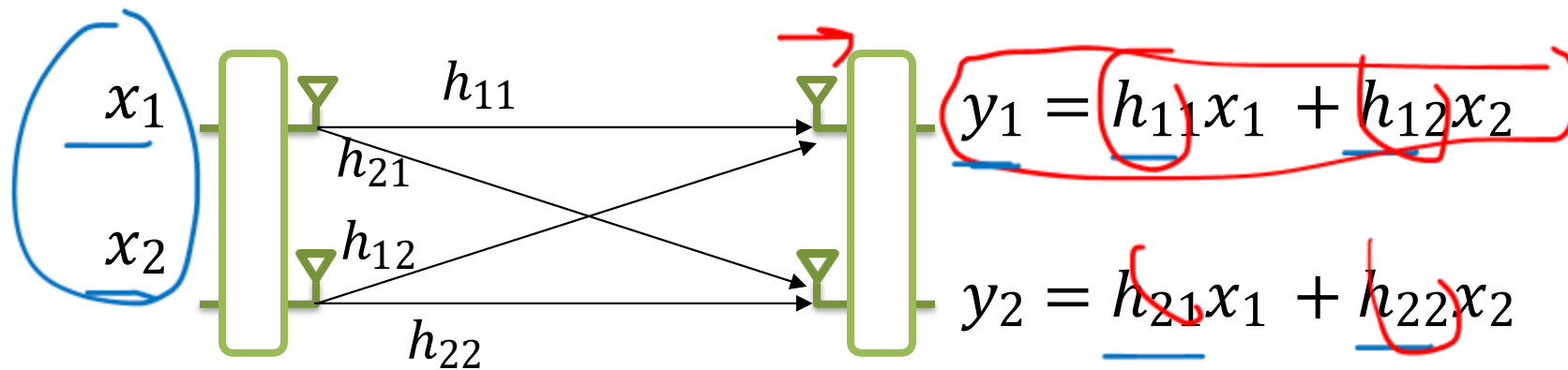
MIMO Channel



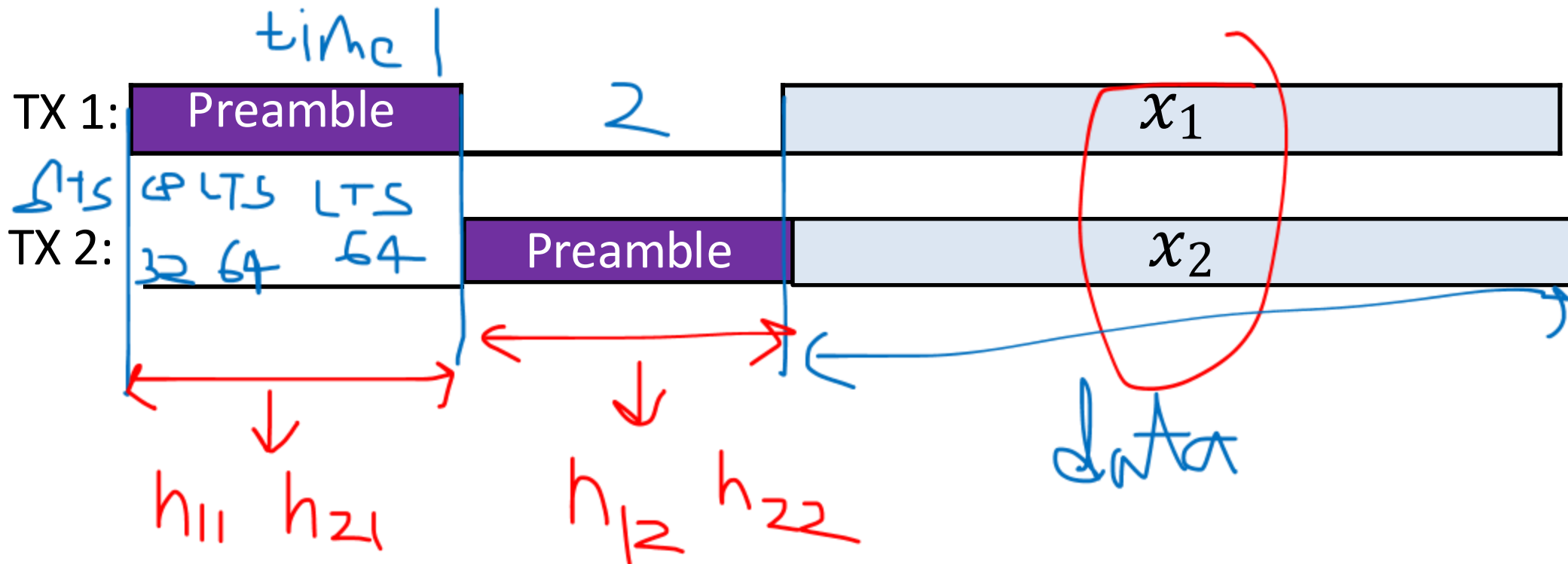
How to estimate the channels: h_{11} , h_{12} , h_{21} , h_{22} ?



MIMO Channel



How to estimate the channels: h_{11} , h_{12} , h_{21} , h_{22} ?



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 - **Receiver Diversity**
 - **Transmitter Diversity**
- 

MIMO Gains

Multiplexing Gain:

- Send multiple packets at the same time
- $N \times N$ MIMO $\rightarrow$ $N \times$ more packets

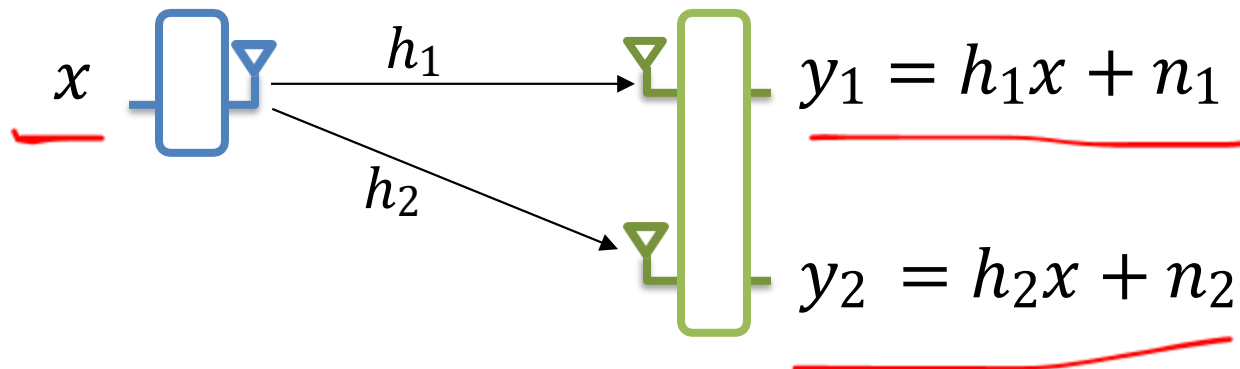
Diversity Gain:

- Send/Receive the same packet on multiple antennas
- Increase SNR of the received packets
 $\rightarrow$ transmit at higher data rates

SNR $\uparrow$ MCS $\uparrow$

Receiver Diversity

51 M0



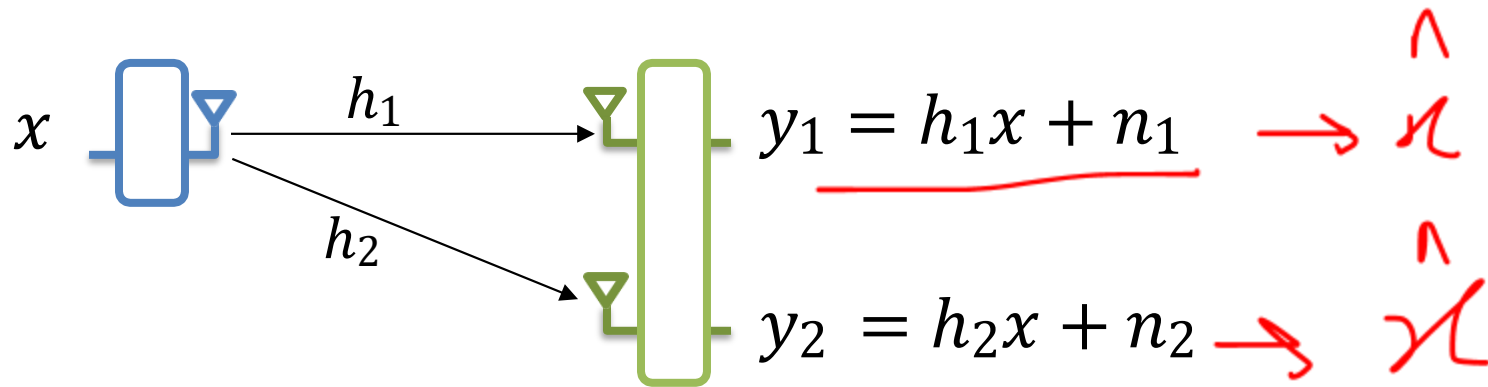
How to best decode x ?

Option 1: Add the received signals

$$\begin{aligned} y_1 + y_2 &= h_1x + n_1 + h_2x + n_2 \\ \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \\ 2 &= (h_1 + h_2)x + n_1 + n_2 \\ \quad \underline{\quad} \quad \underline{\quad} \end{aligned}$$

Channels can sum up destructively! $h_1 + h_2 \approx 0$

Receiver Diversity



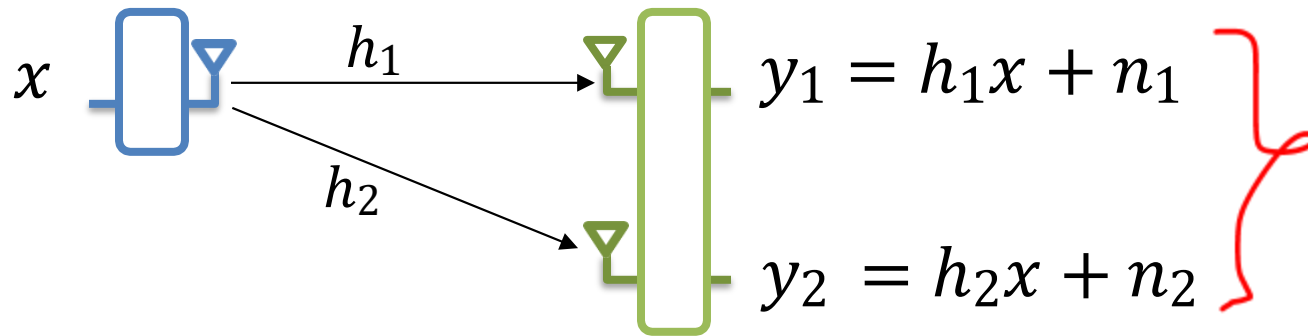
How to best decode x ?

Option 1: Add the received signals

Option 2: ~~Decode independently~~

Sub-optimal!

Receiver Diversity



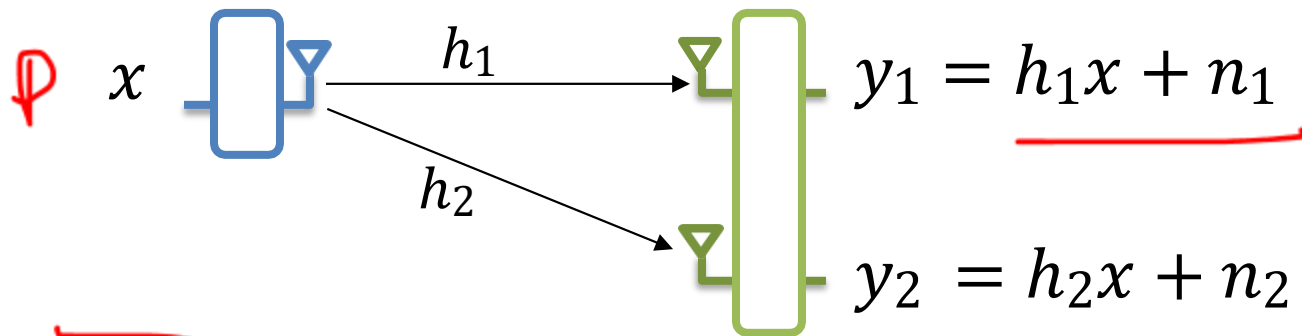
How to best decode x ?

Option 1: Add the received signals

Option 2: Decode independently

Optimal Solution: Maximum Ratio Combining (MRC)

Maximum Ratio Combining

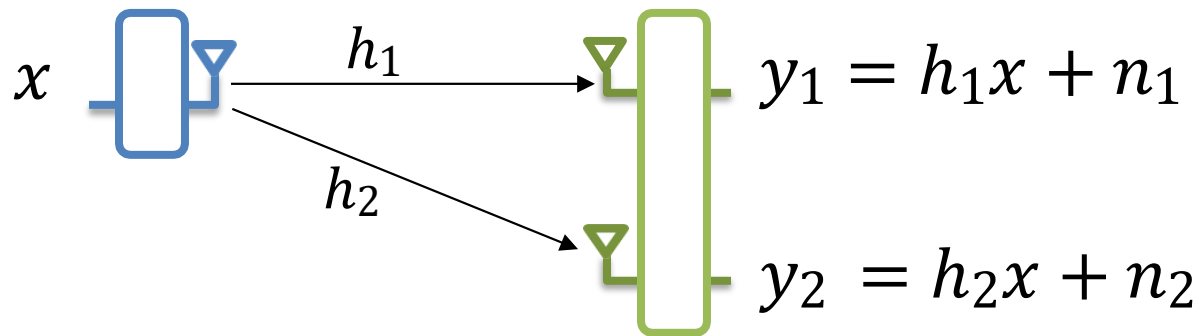


$$\begin{aligned} \left| \alpha_1 y_1 + \alpha_2 y_2 \right|^2 &= h_1^* y_1 + h_2^* y_2 = \underbrace{h_1^* h_1 x + h_1^* n_1}_{\text{Signal}} + \underbrace{h_2^* h_2 x + h_2^* n_2}_{\text{Noise}} \\ &= \underbrace{(|h_1|^2 + |h_2|^2)}_{\text{Total Gain}} x + \underbrace{h_1^* n_1 + h_2^* n_2}_{\text{Noise}} \end{aligned}$$

Let $P = E[|x|^2]$ and $\sigma^2 = E[|n_1|^2] = E[|n_2|^2]$

$$\begin{aligned} \text{Signal Power} &= E[|(|h_1|^2 + |h_2|^2)x|^2] \\ &= (|h_1|^2 + |h_2|^2)^2 E[|x|^2] \\ &= (|h_1|^2 + |h_2|^2)^2 P \end{aligned}$$

Maximum Ratio Combining



$$\begin{aligned} \alpha_1 y_1 + \alpha_2 y_2 &= h_1^* y_1 + h_2^* y_2 = h_1^* h_1 x + h_1^* n_1 + h_2^* h_2 x + h_2^* n_2 \\ &= (|h_1|^2 + |h_2|^2)x + \underline{h_1^* n_1 + h_2^* n_2} \end{aligned}$$

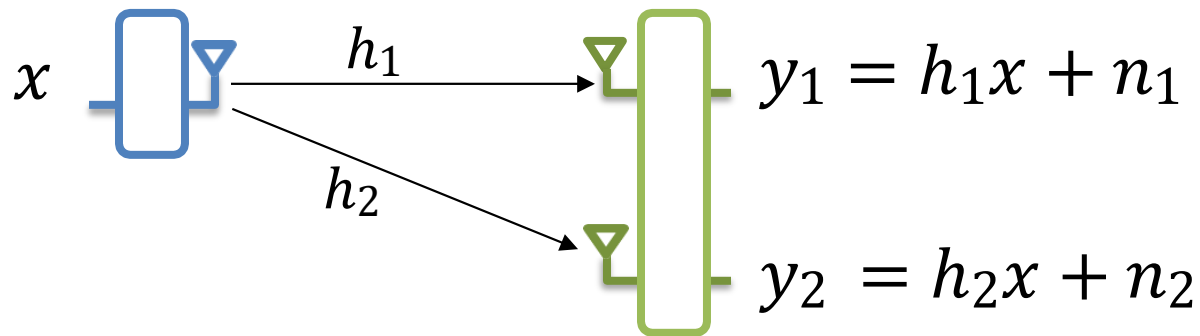
Let $P = E[|x|^2]$ and $\sigma^2 = E[|n_1|^2] = E[|n_2|^2]$

$$\text{Signal Power} = \underline{(|h_1|^2 + |h_2|^2)^2 P}$$

$$\underline{\text{Noise Power}} = E[|h_1^* n_1 + h_2^* n_2|^2] = E[|h_1^* n_1|^2] + E[|h_2^* n_2|^2]$$

$$= \underline{|h_1|^2 E[|n_1|^2]} + \underline{|h_2|^2 E[|n_2|^2]} = \underline{(|h_1|^2 + |h_2|^2) \sigma^2}$$

Maximum Ratio Combining



$$\begin{aligned}\alpha_1 y_1 + \alpha_2 y_2 &= h_1^* y_1 + h_2^* y_2 = h_1^* h_1 x + h_1^* n_1 + h_2^* h_2 x + h_2^* n_2 \\ &= (|h_1|^2 + |h_2|^2)x + h_1^* n_1 + h_2^* n_2\end{aligned}$$

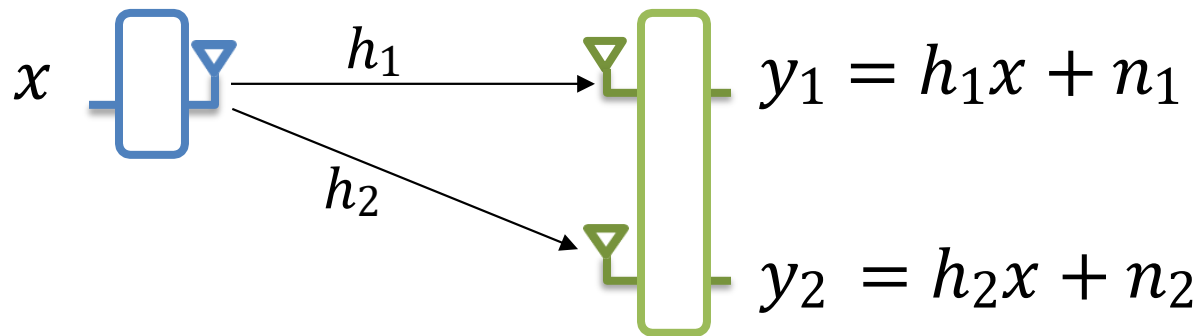
Let $P = E[|x|^2]$ and $\sigma^2 = E[|n_1|^2] = E[|n_2|^2]$

$$\text{Signal Power} = (|h_1|^2 + |h_2|^2)^2 P$$

$$\text{Noise Power} = (|h_1|^2 + |h_2|^2)\sigma^2$$

$$SNR = \frac{(|h_1|^2 + |h_2|^2)^2 P}{(|h_1|^2 + |h_2|^2)\sigma^2} = \underbrace{(|h_1|^2 + |h_2|^2)}_{\text{MRC Gain}} \boxed{\frac{P}{\sigma^2}}$$

Maximum Ratio Combining



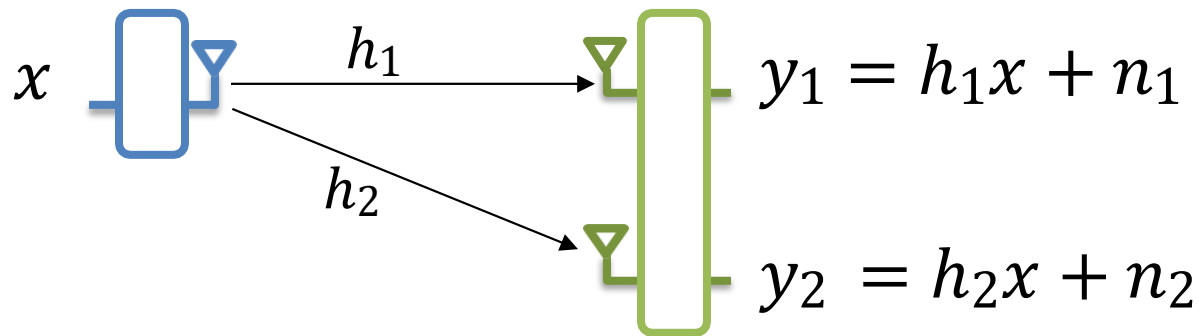
$$\begin{aligned} \alpha_1 y_1 + \alpha_2 y_2 &= h_1^* y_1 + h_2^* y_2 = h_1^* h_1 x + h_1^* n_1 + h_2^* h_2 x + h_2^* n_2 \\ &= (|h_1|^2 + |h_2|^2) x + h_1^* n_1 + h_2^* n_2 \end{aligned}$$

With Receiver Diversity: $SNR = (|h_1|^2 + |h_2|^2) \frac{P}{\sigma^2}$

MRC

Single Receiver: $SNR = |h_1|^2 \frac{P}{\sigma^2}$

Receiver Diversity Gain



With Receiver Diversity: $SNR = (|h_1|^2 + |h_2|^2) \frac{P}{\sigma^2} = 2 |h_1|^2 \frac{P}{\sigma^2}$

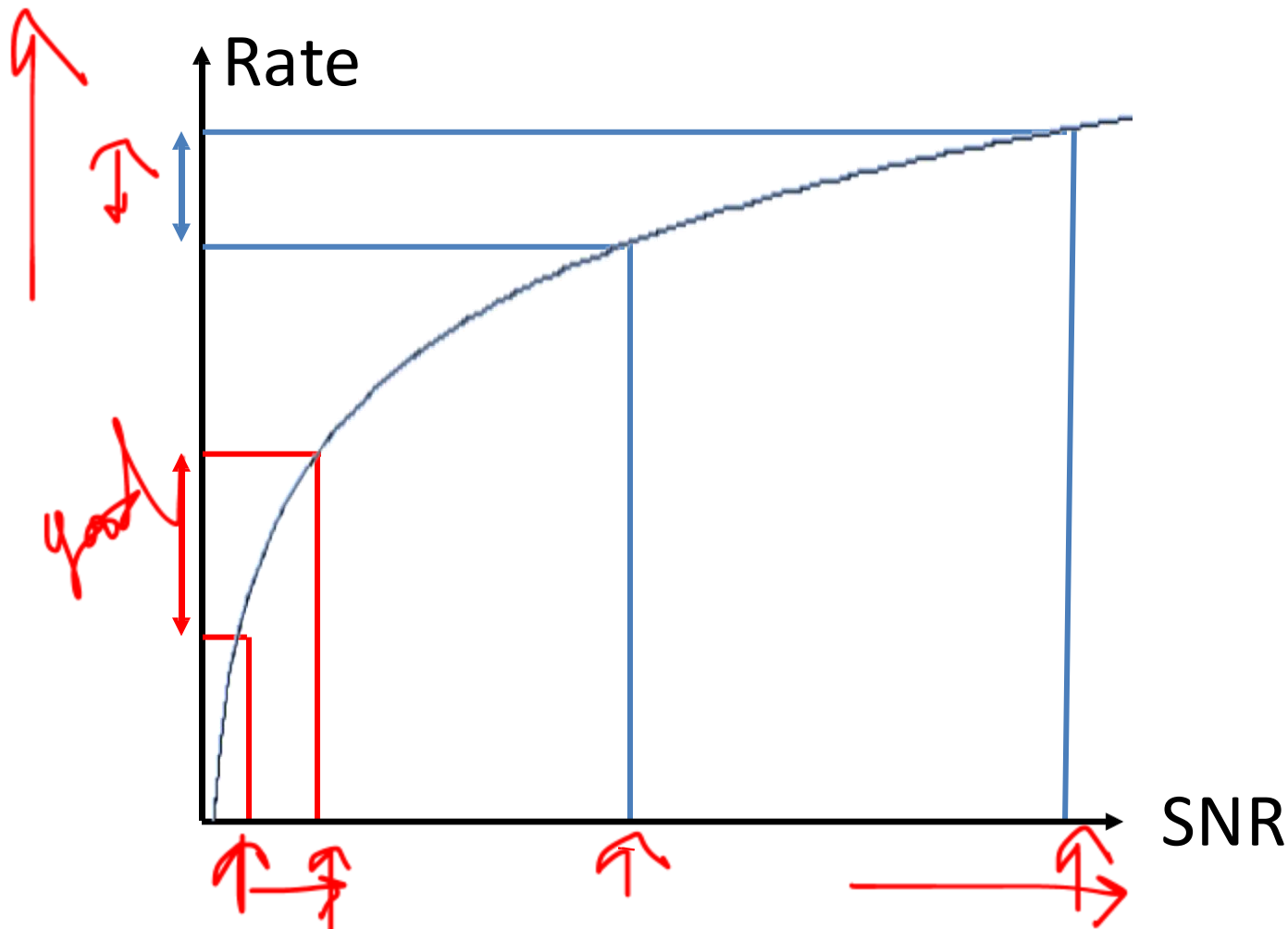
Single Receiver: $SNR = |h_1|^2 \frac{P}{\sigma^2}$

- $|h_1|^2 \approx |h_2|^2$ $\rightarrow$ Can double SNR!

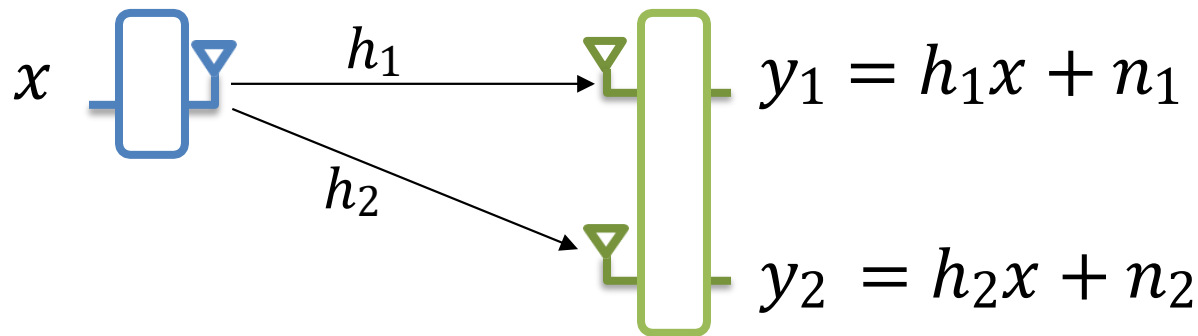
Receiver Diversity Gain

Do we care about doubling the SNR?

$$Capacity \propto \log(SNR)$$



Receiver Diversity Gain



With Receiver Diversity: $SNR_{2R}) = (|h_1|^2 + |h_2|^2) \frac{P}{\sigma^2}$

Single Receiver: $SNR_{1R}) = |h_1|^2 \frac{P}{\sigma^2}$

• $|h_1|^2 \approx |h_2|^2 \rightarrow$ Can double SNR!

• $|h_1|^2 \ll |h_2|^2 \rightarrow$ Huge Gain in SNR

• $|h_1|^2 \gg |h_2|^2 \rightarrow$ Little Gain in SNR